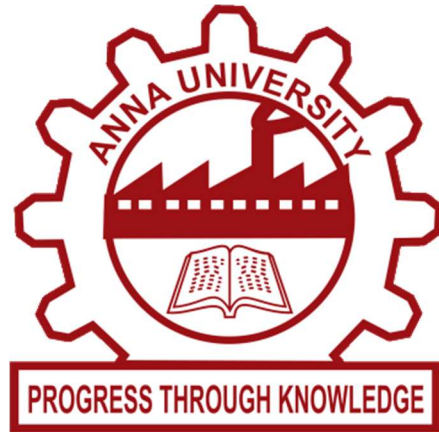




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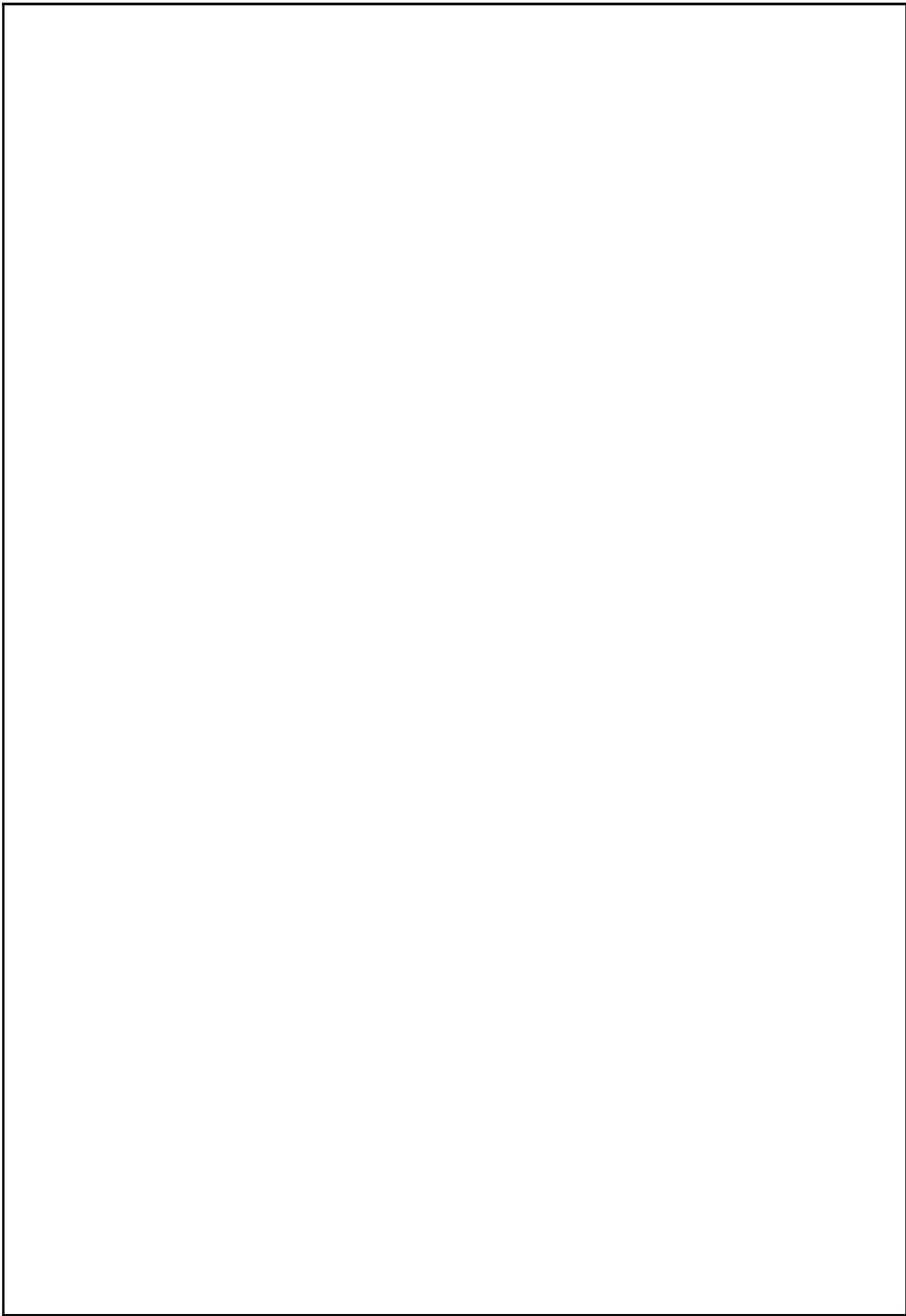
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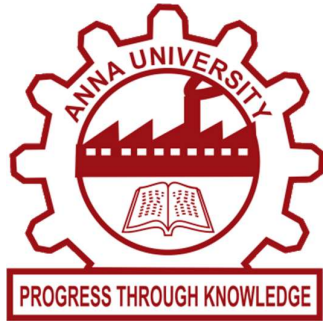
PROJECT REPORT SUBMITTED BY
RUBESH K
AHAMED IBRAHIM AS

In partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING

in

ELECTRONICS AND COMMUNICATION
ENGINEERING





**ANNA UNIVERSITY COLLEGE OF
ENGINEERING, GUINDY**

BONAFIDE CERTIFICATE

Name : RUBESH K , AHAMED IBRAHIM AS

Class : B.E. ECE

Roll No : 2022105702 , 2022105521

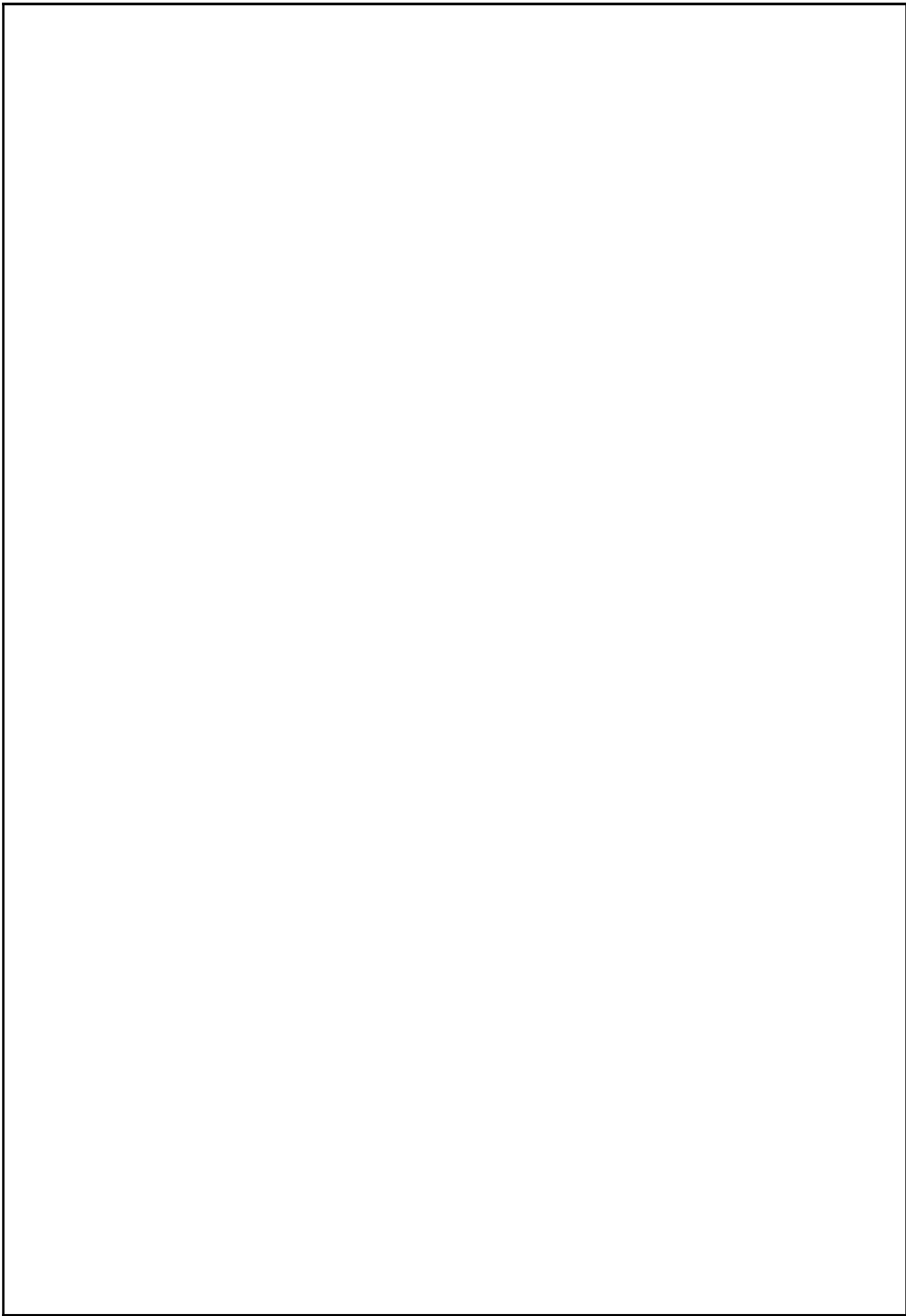
Certified that this is the bonafide record of work done by the
above student in the **VLSI** Laboratory during the session Jan to
May 2025.

*Signature of Lab-in-charge
Department*

Signature of Head of

Submitted for the Practical Examination held on

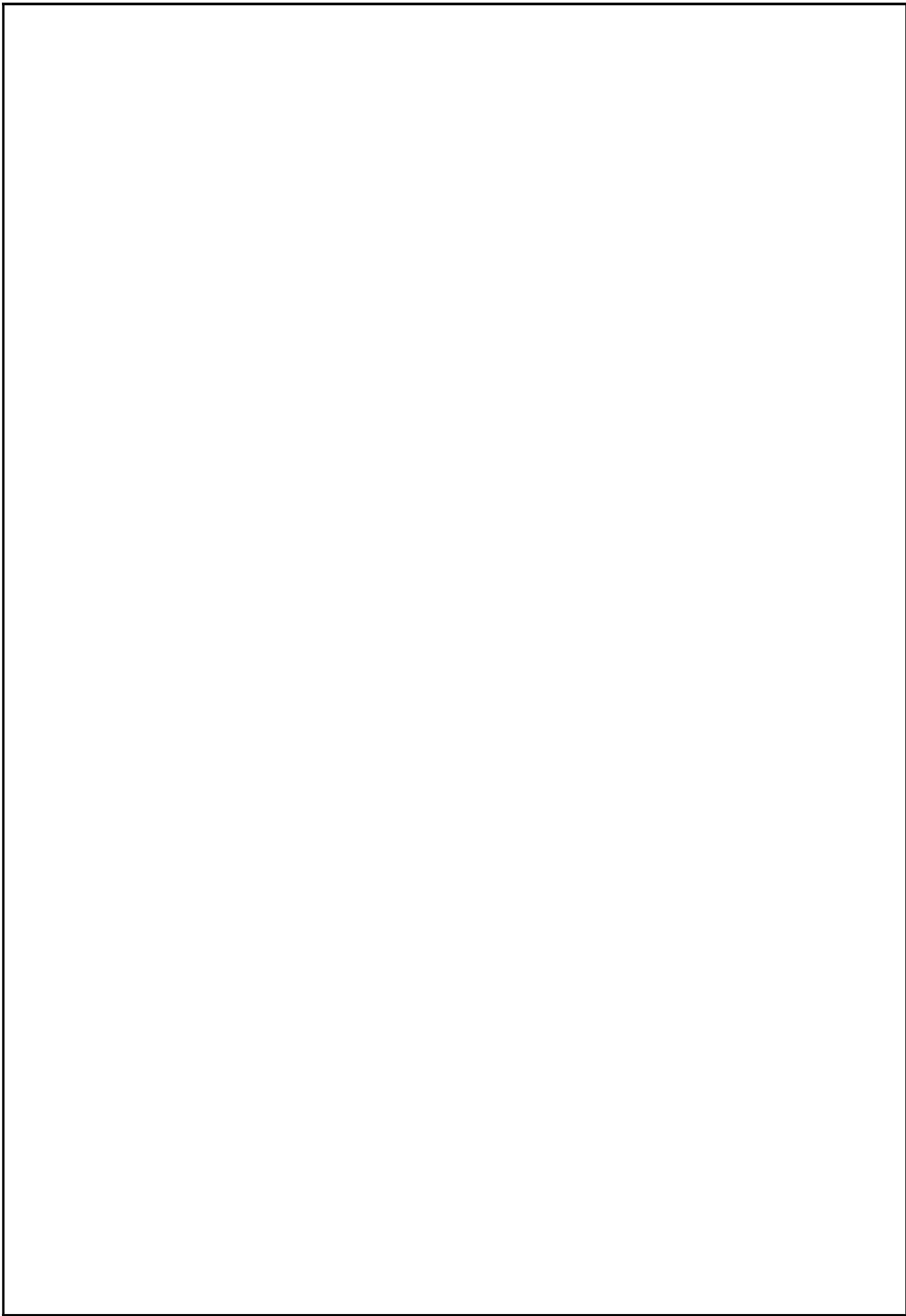
Internal Examiner



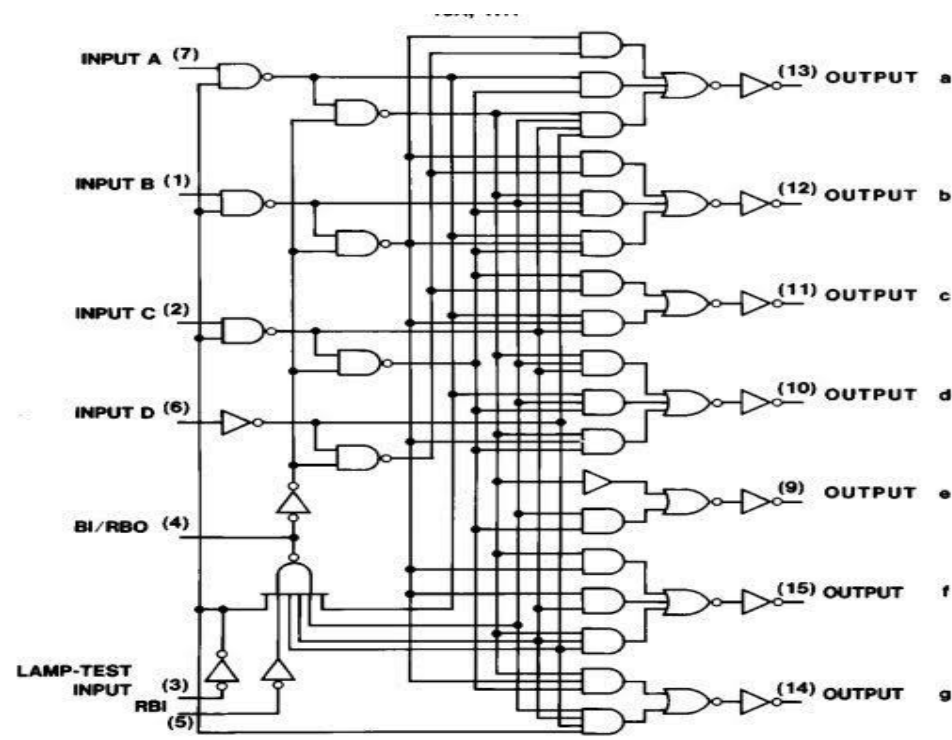
EC5611
VLSI LABORATORY
MINI PROJECT REPORT

DESIGN AND IMPLEMENTATION OF IC 7447 LOGIC
IMPLEMENTATION IN VERILOG

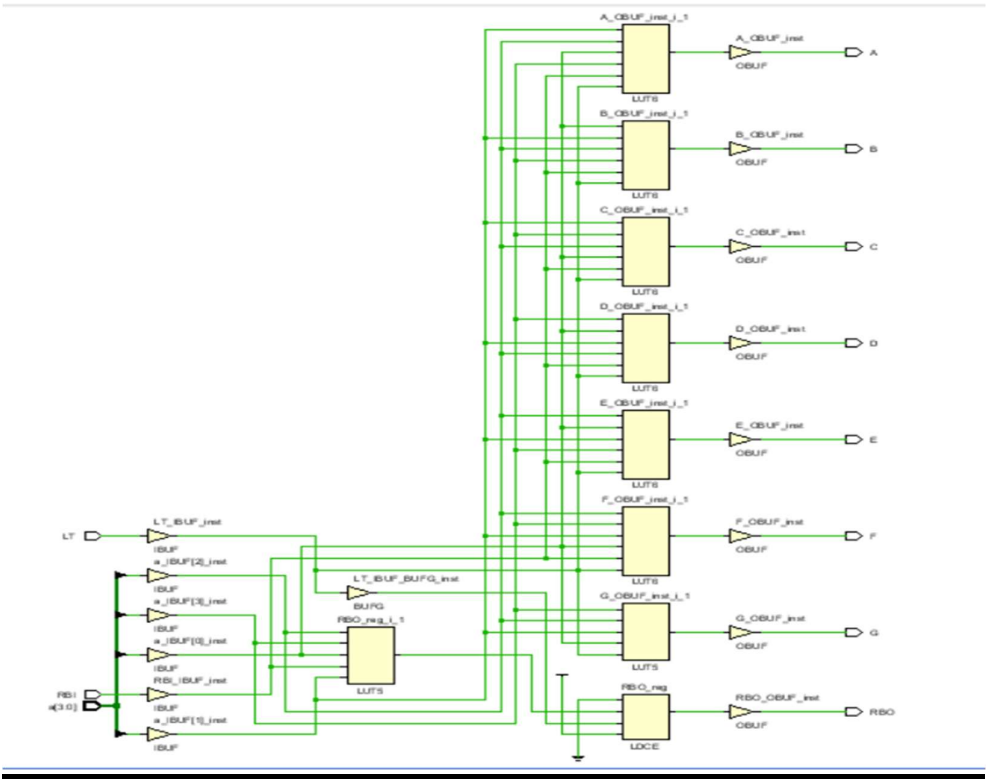
RUBESH K(2022105702)
AHAMED IBRAHIM AS(2022105521)



BLOCK DIAGRAM:



SCHEMATIC DIAGRAM:



DESIGN AND IMPLEMENTATION OF IC 7447

LOGIC IMPLEMENTATION IN VERILOG

AIM:

The objective of this project is to design and implement a digital logic module in Verilog that replicates the functional behavior of the IC 7447 BCD to 7-Segment Decoder/Driver. This module accepts a 4-bit Binary Coded Decimal (BCD) input and generates corresponding outputs to drive a 7-segment display. Additionally, the design incorporates support for control features such as Lamp Test (LT), Ripple Blanking Input (RBI), and Ripple Blanking Output (RBO), enabling accurate simulation of display behavior, including leading zero suppression and test mode activation. The module is validated through a Verilog testbench to simulate the input conditions and observe segment activation.

SOFTWARE REQUIRED:

VIVADO 2017.4

DESCRIPTION:

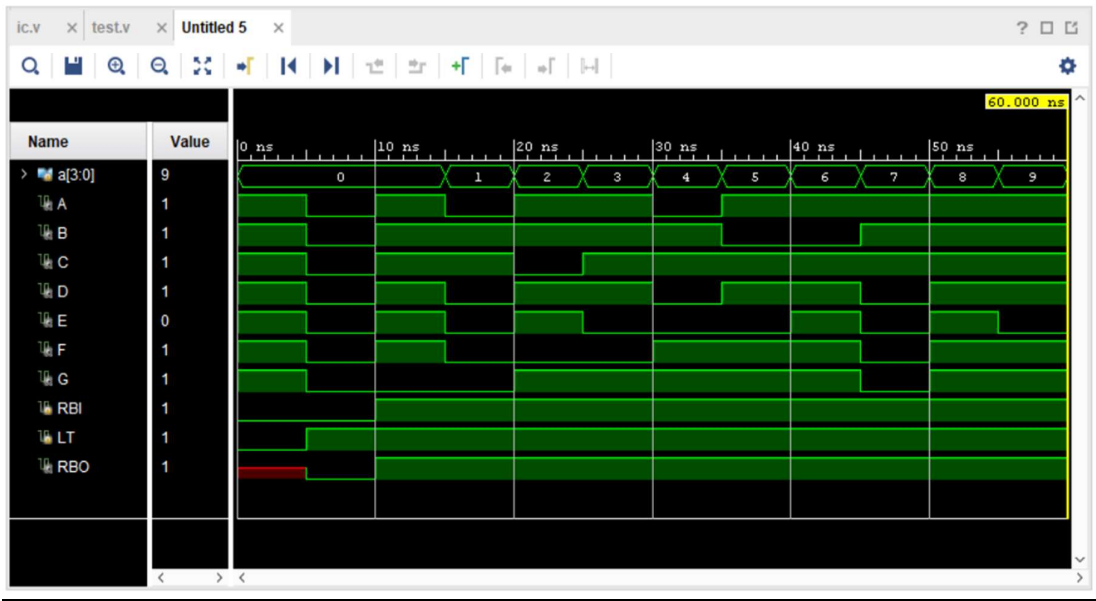
This project focuses on the design and simulation of a digital logic circuit that emulates the functionality of the IC 7447 BCD to 7-Segment Decoder/Driver using Verilog HDL. The IC 7447 is a commonly used decoder in digital systems, especially for driving common-anode 7-segment displays from a BCD (Binary Coded Decimal) input.

The developed Verilog module takes a 4-bit BCD input ($a[3:0]$) and translates it into seven individual output signals (A to G), which correspond to the seven segments of the display. These outputs control which segments light up to visually represent the decimal digit (0–9) on the display.

In addition to the core decoding functionality, the module includes two control inputs and one control output:

- **Lamp Test (LT):** Forces all segments to turn ON regardless of the input, allowing for visual testing of the display hardware.
- **Ripple Blanking Input (RBI):** Used to suppress the display of leading zeros in multi-digit numbers.
- **Ripple Blanking Output (RBO):** Provides an output signal to cascade ripple blanking to additional decoder modules.

WAVEFORM:



POWER:

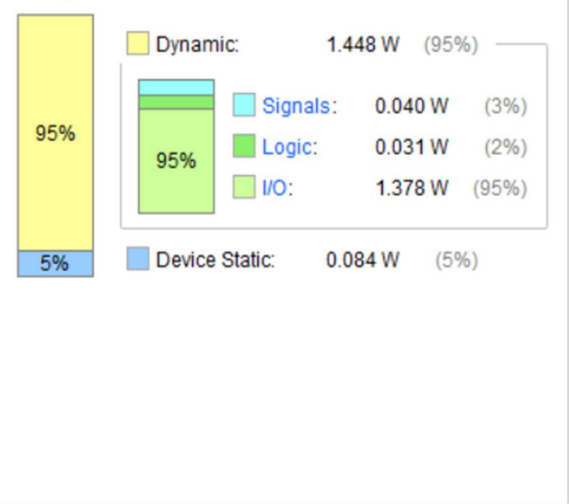
Summary

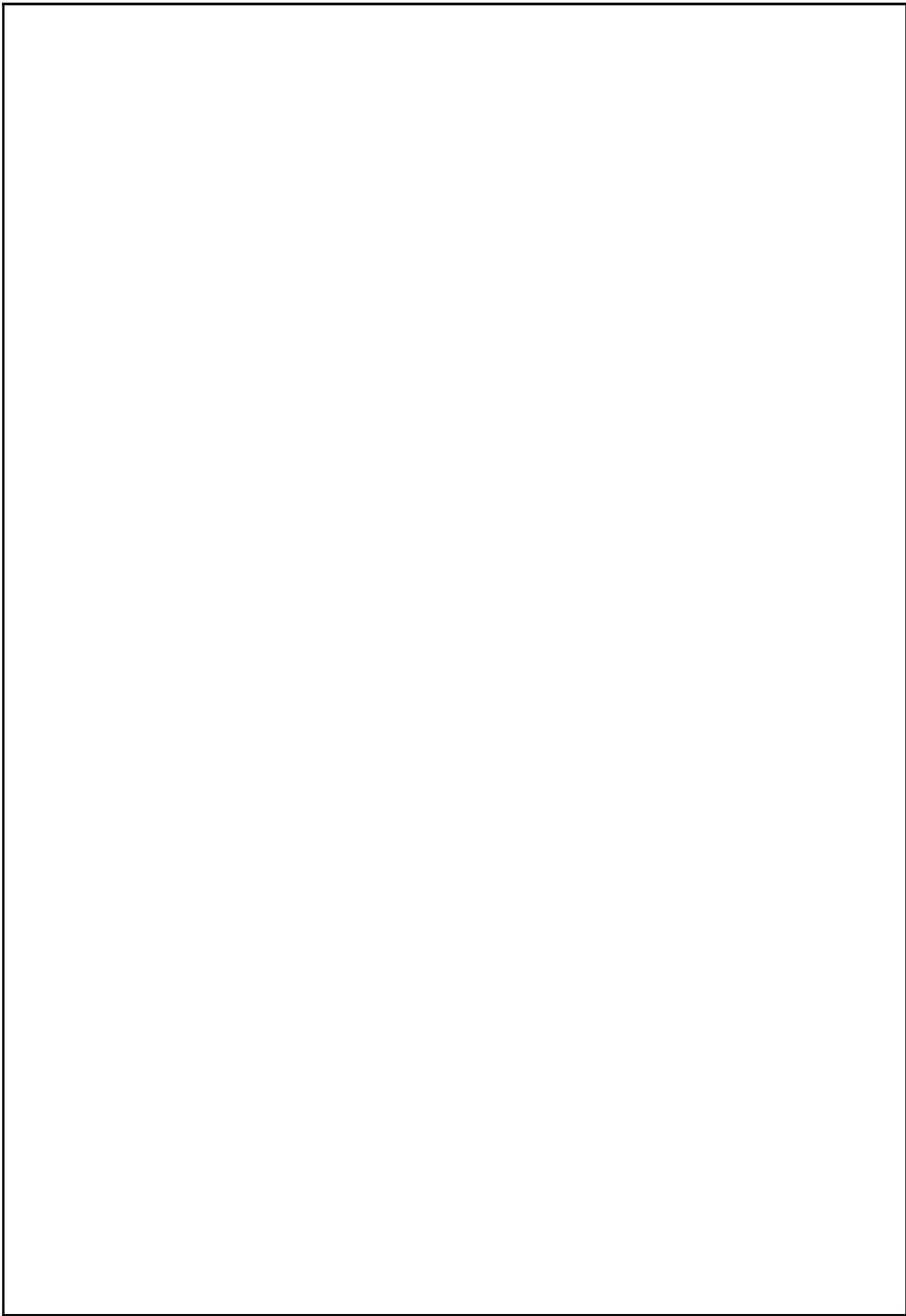
Power estimation from Synthesized netlist. Activity derived from constraints files, simulation files or vectorless analysis. Note: these early estimates can change after implementation.

Total On-Chip Power:	1.532 W
Design Power Budget:	Not Specified
Power Budget Margin:	N/A
Junction Temperature:	27.9°C
Thermal Margin:	57.1°C (30.1 W)
Effective θ_{JA} :	1.9°C/W
Power supplied to off-chip devices:	0 W
Confidence level:	Low

[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

On-Chip Power





PROCEDURE:

The following steps were carried out to design, implement, and verify the IC 7447-based BCD to 7-segment display decoder module using Verilog HDL:

1. Study of IC 7447 Functional Characteristics

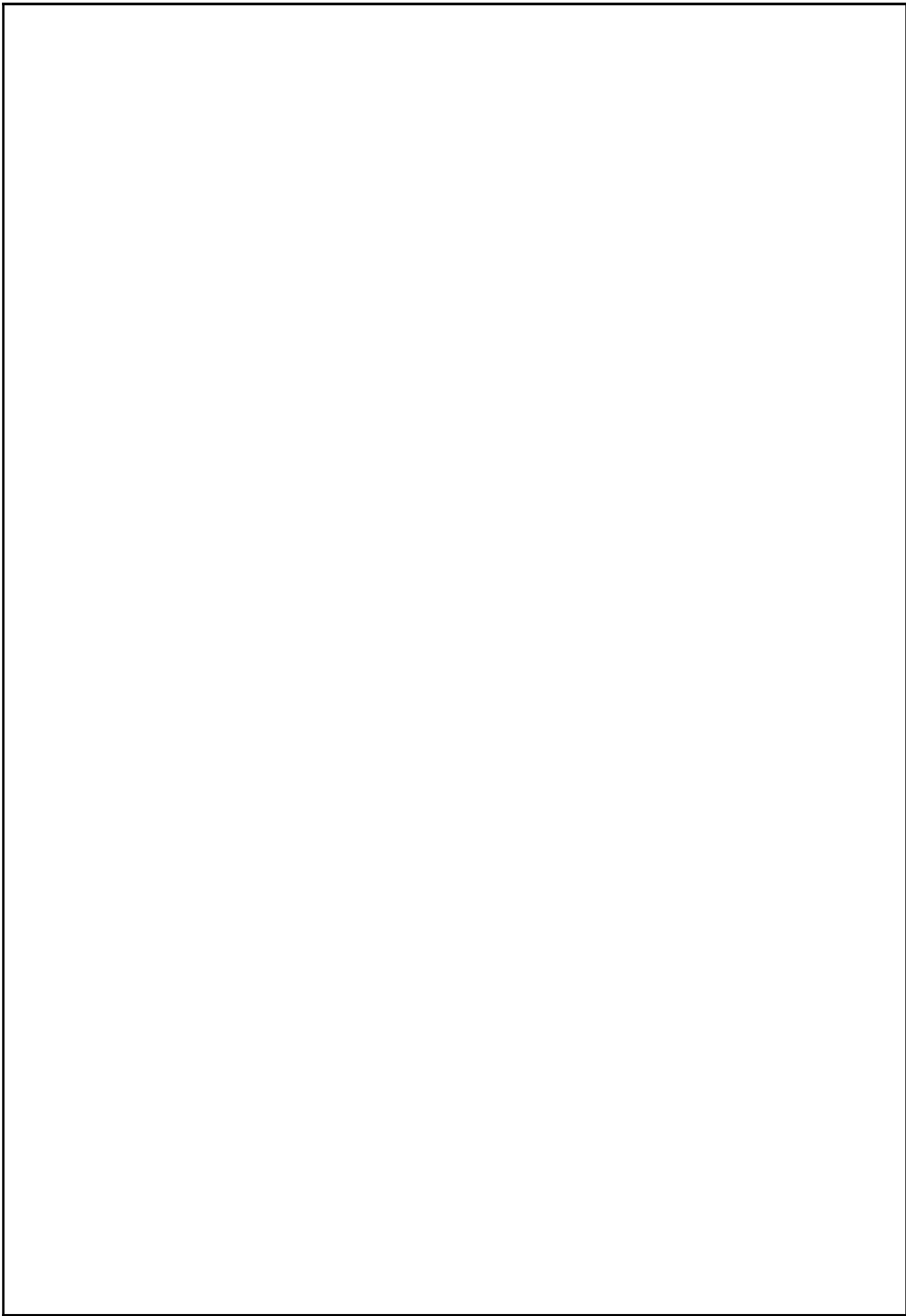
- Referred to the datasheet of the IC 7447 to understand its pin configuration, logic behavior, truth table, and control features like LT, RBI, and RBO.
- Analyzed the logic expressions required to generate segment outputs (A to G) based on 4-bit BCD inputs.

2. Writing the Verilog Module

- Created a Verilog module named ic that:
 - Takes inputs: a[3:0], LT, and RBI
 - Produces outputs: A, B, C, D, E, F, G, and RBO
- Implemented combinational logic using Boolean expressions for each segment.
- Handled control inputs:
 - When $LT = 0$, all segments are turned ON.
 - When $RBI = 0$ and $a = 0000$, all segments are turned OFF and $RBO = 0$.

3. Designing the Testbench

- Wrote a testbench module to simulate and validate the functionality of the design.
- Initialized inputs and applied a sequence of BCD values (0000 to 1001) along with control signals.
- Included delays to observe transitions and simulate real-time behavior.
- Used \$monitor and \$display to observe output segment values during simulation.



4.Simulation

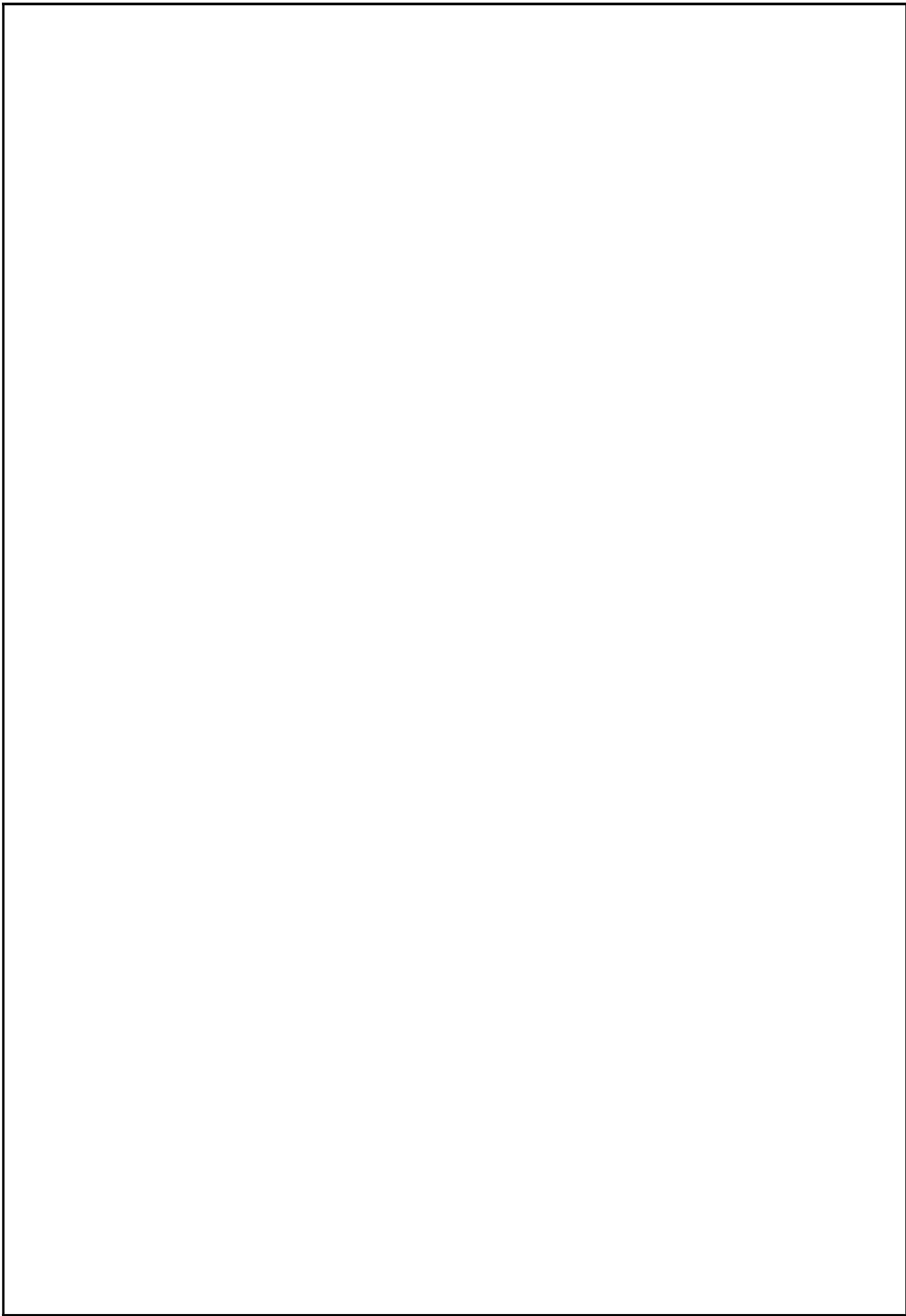
- Compiled the Verilog module and testbench using a Verilog simulator (such as ModelSim, Vivado Simulator, or any other supported tool).
- Verified the correctness of output segment patterns for each BCD input.
- Checked behavior when $LT = 0$ (all segments ON) and when $RBI = 0$ for input 0000.

5.Result analysis

- Compared simulation results with the standard IC 7447 truth table to confirm functional accuracy.
- Verified segment outputs and RBO signal for correctness under all test conditions.

CODE:

```
module ic(
    input [3:0] a,
    output reg A, B, C, D, E, F, G,
    input RBI,LT,
    output reg RBO
);
always @(*) begin
    if (LT ==0) begin
        A=1; B=1; C=1; D=1;E=1;F=1;G=1;
    end
    else if (RBI ==0 && a==0000) begin
        A=0; B=0; C=0; D=0; E=0; F=0; G=0; RBO=0;
    end else begin
        RBO =1;
    end
end
```



```

    A = (a[1] | a[3] | a[2]&a[0] | ~(a[2]|a[0]));
    B = ~a[2] | ~(a[1]|a[0]) | a[1]&a[0] ;
    C = ~a[1] | a[0] | ~a[3]&a[2];
    D=
a[1]&a[0]|a[3]|~a[2]&a[1]|a[2]&~a[1]&a[0]|(a[2]|a[1]|a[0]);
    E = a[1]&~a[0] | ~(a[2]|a[1]|a[0]) ;
    F = a[3] | a[2]&~a[0] | a[2]&~a[1] | ~(a[1]|a[0]);
    G = a[1]&~a[0] | a[1]&~a[2] | a[2]&~a[1] |
a[3] ;
    end

end

endmodule

```

TESTBENCH:

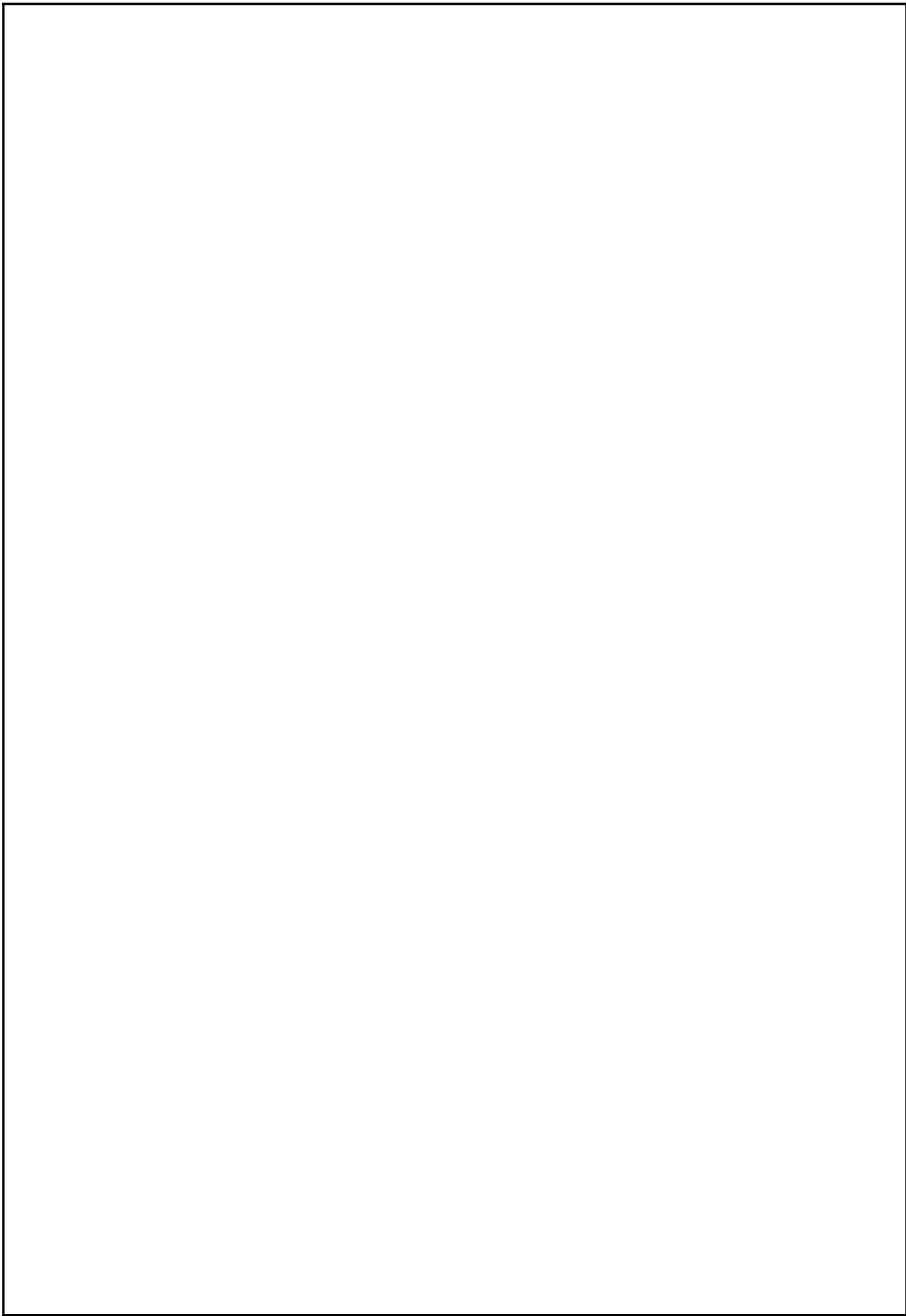
```

module test( );
    reg [3:0] a;
    wire A, B, C, D, E, F, G;
    reg RBI,LT;
    wire RBO;

    ic uut(a,A,B,C,D,E,F,G,RBI,LT,RBO);

    initial begin
        LT =0; a=0; RBI= 0;
        #5; LT=1;
        #5; RBI = 1;
        #5; a=1;
        #5; a=2;
        #5; a=3;
        #5; a=4;
        #5; a=5;
        #5; a=6;
        #5; a=7;
    end

```




```
#5; a=8;  
#5; a=9;  
#5;$finish;  
end
```

```
endmodule
```

RESULT

The Verilog-based simulation of the IC 7447 BCD to 7-segment decoder was successfully implemented and tested using a testbench. The simulation results confirmed that the module correctly generated the appropriate segment outputs (A to G) for all valid BCD inputs (0000 to 1001).